Assessment Menu: Teacher Guide

The Congruence Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
CRITERIA	Choose and justify a congruence criterion	A1, A2, E1, E2, S4
PARTS	Corresponding parts and CPCTC with algebra	A3, A4, E3, E4, S2
ISOSCELES	Isosceles and equilateral triangle theorems	A5, E5, S1
REASON	Reason about what is (and is not) enough	A6, E6, S3

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 CRITERIA	In △ABC and △DEF, AB = DE = 5, BC = EF = 7, and CA = FD = 9. Which congruence criterion applies, and what is the congruence statement?	**Answer:** SSS; △ABC ≅ △DEF All three pairs of sides are congruent, so the criterion is **SSS**. Match vertices in order: A↔D, B↔E, C↔F, so △ABC ≅ △DEF.	2 pts: Names SSS 3 pts: Writes △ABC ≅ △DEF with vertices in matching order
A2 CRITERIA	In △PQR and △XYZ, PQ = XY = 7, ∠Q = ∠Y = 40°, and QR = YZ = 9. Which criterion proves △PQR ≅ △XYZ? Explain why the angle matters.	**Answer:** SAS: ∠Q is the angle between sides PQ and QR (and ∠Y between XY and YZ). Two pairs of sides and the angle **between** them are congruent. ∠Q is formed by PQ and QR, so it is the included angle: **SAS**.	2 pts: Names SAS 3 pts: Explains that the angle is between (included by) the two sides
A3 PARTS	△ABC ≅ △DEF with ∠A = 50° and ∠B = 60°. Find ∠F and name the side congruent to AC.	**Answer:** ∠F = 70°; AC ≅ DF ∠C = 180° − 50° − 60° = 70°. F corresponds to C, so ∠F = 70°. A↔D and C↔F, so AC ≅ DF.	2 pts: Finds ∠C = 70° 1 pts: Uses correspondence: ∠F = 70° 2 pts: Names DF
A4 PARTS	△JKL ≅ △MNP with JK = 3x + 2 and MN = 20. Find x and JK.	**Answer:** x = 6, JK = 20 JK and MN are corresponding sides, so they are equal: 3x + 2 = 20. 3x = 18, so x = 6. JK = 3(6) + 2 = 20.	2 pts: Sets 3x + 2 = 20 2 pts: Solves x = 6 1 pts: JK = 20
A5 ISOSCELES	△ABC is equilateral with AB = 2x + 1 and BC = x + 7. Find x and the measure of each angle.	**Answer:** x = 6; each angle is 60° All sides are equal: 2x + 1 = x + 7, so x = 6 (each side is 13). An equilateral triangle is also equiangular, so each angle is 180° ÷ 3 = 60°.	2 pts: Sets 2x + 1 = x + 7 and solves x = 6 3 pts: States each angle is 60° (equiangular)
A6 REASON	Two triangles each have angles 40°, 60°, and 80°. A student says they must be congruent. Is the student right? Explain.	**Answer:** No. AAA does not guarantee congruence: the triangles can have the same shape but different sizes. Equal angles only guarantee the same **shape** (similar), not the same size. Example: a triangle and a copy with every side doubled have identical angles but are not congruent.	2 pts: Answers “No” 3 pts: Explains that AAA allows different sizes (gives a size-change example)

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 CRITERIA	△ABC has ∠B = 90°, AB = 5, AC = 13. △DEF has ∠E = 90°, DE = 5, DF = 13. Show the triangles are congruent, naming the criterion, then find EF.	**Answer:** HL; △ABC ≅ △DEF; EF = BC = 12 Both are right triangles with congruent hypotenuses (AC = DF = 13) and a congruent leg (AB = DE = 5): **HL**. So △ABC ≅ △DEF. BC = $\sqrt{(13^2 - 5^2)}$ = √144 = 12. EF corresponds to BC (CPCTC), so EF = 12.	3 pts: Names HL and identifies right angles, congruent hypotenuses and a congruent leg 2 pts: Correct statement △ABC ≅ △DEF 3 pts: Finds BC = 12 with the Pythagorean theorem 2 pts: EF = 12 by CPCTC
E2 CRITERIA	In △ABC and △DEF, ∠A = ∠D = 45°, ∠B = ∠E = 70°, and BC = EF = 8. Find ∠C and ∠F. Which criterion applies directly to the given information, and why is it not ASA?	**Answer:** ∠C = ∠F = 65°; AAS, because BC is not between ∠A and ∠B. ∠C = 180° − 45° − 70° = 65° and ∠F = 180° − 45° − 70° = 65°. The given side BC touches ∠B and ∠C, not the two given angles ∠A and ∠B. So it is **AAS**, not ASA. So △ABC ≅ △DEF by AAS.	2 pts: ∠C = ∠F = 65° 3 pts: Names AAS 3 pts: Explains the side is not between the two given angles 2 pts: Correct congruence statement △ABC ≅ △DEF
E3 PARTS	△ABC ≅ △DEF. AB = 2x + 3, DE = 5x − 12, BC = 4y − 1, and EF = y + 11. Find x, y, AB, and BC.	**Answer:** x = 5, y = 4, AB = 13, BC = 15 AB = DE: 2x + 3 = 5x − 12, so 15 = 3x and x = 5. AB = 2(5) + 3 = 13. BC = EF: 4y − 1 = y + 11, so 3y = 12 and y = 4. BC = 4(4) − 1 = 15.	2 pts: Sets AB = DE and solves x = 5 2 pts: Sets BC = EF and solves y = 4 3 pts: AB = 13 and BC = 15 3 pts: Correct correspondence AB↔DE and BC↔EF (shown by the equations)
E4 PARTS	△ABC ≅ △DEF with ∠A = (3x + 5)°, ∠D = (5x − 15)°, and ∠B = 70°. Find x, then ∠C and ∠F.	**Answer:** x = 10, ∠C = 75°, ∠F = 75° ∠A = ∠D: 3x + 5 = 5x − 15, so x = 10. ∠A = 3(10) + 5 = 35°. ∠C = 180° − 35° − 70° = 75°. ∠F corresponds to ∠C, so ∠F = 75°.	3 pts: Sets 3x + 5 = 5x − 15 2 pts: Solves x = 10 3 pts: Finds ∠C = 75° 2 pts: ∠F = 75° by CPCTC
E5 ISOSCELES	△ABC is isosceles with AB = AC. AB = 4x − 3, AC = 2x + 9, and BC = 3x. Find x and the perimeter of △ABC.	**Answer:** x = 6; perimeter = 60 AB = AC: 4x − 3 = 2x + 9, so 2x = 12 and x = 6. AB = AC = 4(6) − 3 = 21 and BC = 3(6) = 18. Perimeter = 21 + 21 + 18 = 60.	3 pts: Sets 4x − 3 = 2x + 9 2 pts: x = 6 2 pts: Side lengths 21, 21, 18 3 pts: Perimeter 60
E6 REASON	Segments AB and CD intersect at M, and M is the midpoint of both. Write a short proof (statements and reasons) that △AMC ≅ △BMD, naming the criterion.	**Answer:** AM = BM (midpoint), ∠AMC = ∠BMD (vertical angles), CM = DM (midpoint); △AMC ≅ △BMD by SAS. AM = BM because M is the midpoint of AB. ∠AMC = ∠BMD because vertical angles are congruent. CM = DM because M is the midpoint of CD.	2 pts: AM = BM with reason (midpoint) 2 pts: CM = DM with reason (midpoint) 3 pts: ∠AMC = ∠BMD with reason (vertical angles) 3 pts: Concludes SAS and writes △AMC ≅ △BMD

		Two sides and the included angle are congruent, so △AMC ≅ △BMD by **SAS**.	

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 ISOSCELES	$\triangle ABC$ has $AB = AC = 13$ and $BC = 10$. D is the midpoint of BC. (a) Prove $\triangle ABD \cong \triangle ACD$, naming the criterion. (b) Use CPCTC to show $AD \perp BC$. (c) Find AD.	**Answer:** (a) SSS (b) $\angle ADB = \angle ADC$ and they are a linear pair, so each is 90° (c) AD = 12 (a) AB = AC (given); BD = CD (D is the midpoint); AD = AD (reflexive). So $\triangle ABD \cong \triangle ACD$ by **SSS**. (b) $\angle ADB = \angle ADC$ by CPCTC. They form a linear pair, so they add to 180°. Each is 90°, so $AD \perp BC$. (c) $BD = 10 \div 2 = 5$. In right $\triangle ABD$: $AD = \sqrt{(13^2 - 5^2)} = \sqrt{144} = 12$.	5 pts: (a) Given and midpoint pairs with reasons (3), AD = AD reflexive (2) 3 pts: (a) Concludes SSS with a correct congruence statement 3 pts: (b) $\angle ADB = \angle ADC$ by CPCTC 3 pts: (b) Linear pair adds to 180°, so each angle is 90° 6 pts: (c) BD = 5 (2), Pythagorean theorem to get AD = 12 (4)
S2 PARTS	$\triangle ABC \cong \triangle DEF$. $AB = 2x + y$, $DE = 17$, $BC = x + 2y$, $EF = 16$, and $AC = 3x + 1$. (a) Find x and y. (b) Find DF. (c) Find the perimeter of $\triangle DEF$.	**Answer:** (a) $x = 6$, $y = 5$ (b) DF = 19 (c) 52 (a) AB = DE gives $2x + y = 17$. BC = EF gives $x + 2y = 16$. Add: $3x + 3y = 33$, so $x + y = 11$. Subtract: $x - y = 1$. So $x = 6$ and $y = 5$. (b) $AC = 3(6) + 1 = 19$, and DF corresponds to AC (CPCTC), so DF = 19. (c) Perimeter of $\triangle DEF = 17 + 16 + 19 = 52$.	6 pts: Writes $2x + y = 17$ and $x + 2y = 16$ using the correct correspondence 4 pts: Solves the system: $x = 6$ (2) and $y = 5$ (2) 6 pts: AC = 19 (3), DF = 19 by CPCTC (3) 4 pts: Perimeter 52 (sides 17, 16, 19)
S3 REASON	In $\triangle ABC$ and $\triangle DEF$, $AB = DE = 8$ and $\angle A = \angle D = 35°$. For each extra fact, say what it proves or that it is not enough, and why: (a) AC = DF (b) $\angle B = \angle E$ (c) BC = EF (d) $\angle C = \angle F$	**Answer:** (a) SAS (b) ASA (c) not enough (SSA) (d) AAS (a) AB, $\angle A$, AC: the angle is between the two sides, so **SAS**. (b) $\angle A$, AB, $\angle B$: the side is between the two angles, so **ASA**. (c) Two sides and a **non-included** angle (SSA) is not a criterion: for some lengths of BC two different triangles fit. **Not enough.** (d) Two angles and a non-included side (AB is opposite $\angle C$): **AAS**.	5 pts: (a) SAS, noting the angle is included between the sides 5 pts: (b) ASA, noting the side is between the angles 5 pts: (c) Not enough: SSA, the angle is not between the sides 5 pts: (d) AAS, noting the side is not between the angles
S4 CRITERIA	In quadrilateral ABCD, $\angle B = \angle D = 90°$ and $AB = AD = 8$. Diagonal $AC = 17$. (a) Prove $\triangle ABC \cong \triangle ADC$, naming the criterion. (b) Use CPCTC to explain why AC bisects $\angle BAD$. (c) Find the perimeter of ABCD.	**Answer:** (a) HL (b) $\angle BAC = \angle DAC$ (c) 46 (a) $\triangle ABC$ and $\triangle ADC$ are right triangles ($\angle B = \angle D = 90°$). Hypotenuse AC = AC (reflexive) and leg AB = AD (given). So they are congruent by **HL**. (b) $\angle BAC = \angle DAC$ by CPCTC, so AC splits $\angle BAD$ into two equal angles: it bisects it. (c) $BC = \sqrt{(17^2 - 8^2)} = \sqrt{225} = 15$. CD = BC = 15 by CPCTC, and AD = 8. Perimeter $= 8 + 15 + 15 + 8 = 46$.	6 pts: (a) Right triangles (2), AC = AC reflexive (2), AB = AD given (2) 2 pts: (a) Concludes HL 4 pts: (b) $\angle BAC = \angle DAC$ by CPCTC (2), links to the definition of angle bisector (2) 8 pts: (c) BC = 15 by Pythagorean theorem (2), CD = 15 by CPCTC (2), perimeter 46 (4)